

Modeling Song Popularity: Neural vs Bayesian

SiyuanChen, Yifan Wang

Introduction

Music streaming platforms like Spotify generate large amounts of data that can be used to study song popularity. In this project, we explore whether audio features such as danceability, energy, tempo, loudness, and valence can be used to predict how popular a song will be.

Background, Motivation and Goals

- Can audio features predict song popularity?
- Understanding popularity helps music platforms and artists
- Compare Neural Network and Bayesian models

Data Description

Dataset: Spotify Tracks Dataset (~114,000 songs)

Features: danceability, energy, tempo, loudness, valence

Target: Popularity (0-100)

Bayesian Regression (R)

Linear model with priors

Posterior estimation

Provides interpretability

Neural Network (NumPy) Architecture: $5 \rightarrow 64 \rightarrow 32 \rightarrow 1$.

ReLU + Linear output.

Mini-batch SGD

Model Comparison and Limitations

Neural Network focuses on prediction performance but shows limited accuracy.

Bayesian regression provides clearer insights into how each feature affects popularity.

However, The model only uses audio features and does not include external factors such as artist popularity, social media influence, or playlist exposure.

Overall, the neural network is better for prediction, while the Bayesian model is better for interpretation.

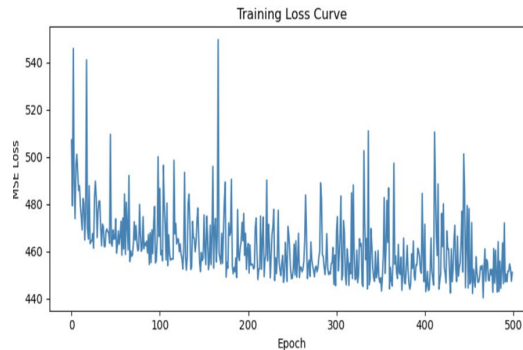


Figure 1: Neural Network Training Loss

Findings:

- Predictions are concentrated around the mean, indicating regression toward the average
- The neural network shows limited predictive performance (low R^2)
- The Bayesian model provides better interpretability of feature effects
- Loudness and danceability have positive effects on popularity
- Audio features alone are not sufficient to fully explain song popularity

Conclusion

Both models suggest that audio features have limited predictive power, but provide useful insights into song characteristics. Future work could include incorporating external factors such as artist popularity, social media engagement, or playlist exposure.

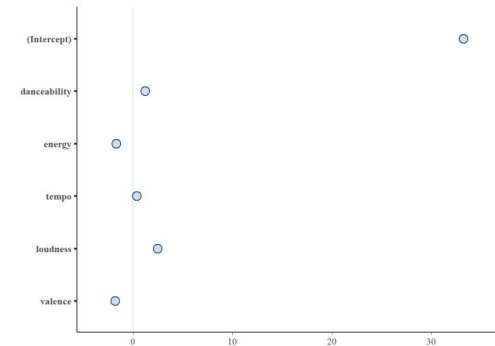


Figure 2: Bayesian Feature Effects on Popularity

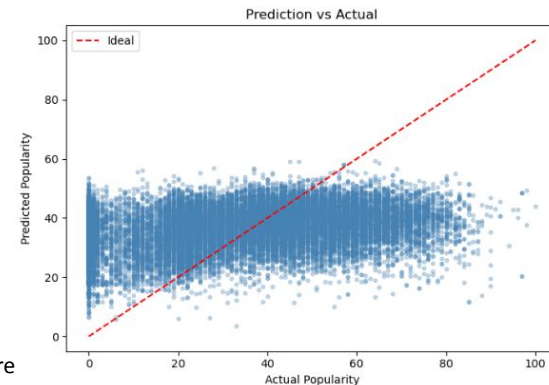


Figure 3: Predicted vs Actual Song Popularity